Fall 2023 / EC571

Instructor:Ajay Joshi

Mid-Term Exam 3

Date: December 11, 2023

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 1 hour 45 minutes.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
6. Material sharing is not allowed.
7. Calculators are allowed.
8. The exam is closed book close notes, but you are allowed to have two cheat sheets.
9. Boston University Academic Code of Conduct applies.
10. BU ENG is collaborating with several online services, including Chegg, to monitor activity during exams and obtain account information and IP addresses connected to uploaded exam materials and their access. The penalties for cheating are extremely severe including expulsion from BU.

Question 1

You recently designed a microprocessor in a 65 *nm* technology with V_{DD} = 1.2 V and V_{TH} = 0.35 V that runs at 2 GHz and consumes a total power of 1 W, where 750 mW is dynamic power and 250 mW is due to subthreshold leakage. In this problem, we are looking at the choices available to us if we were to scale the design to a 45 *nm* technology. Throughout the problem, you should assume that the leakage current is modeled by $W \cdot I_0 \cdot e^{\frac{-V_{TH}}{38mV}}$, where I_0 is the same in both the 65 *nm* and 45 *nm* technologies.

(A) Using the full scaling model and assuming velocity-saturated devices, what would be the new operating frequency, supply voltage, and power consumption of the chip in the 45 *nm* technology? (10 points)

(B) Now let's assume that in moving to the 45 *nm* technology we leave $V_{DD} = 1.2\ V$ and $V_{TH} = 0.35\ V$. Still with velocity-saturated devices and assuming that $\xi_{crit} = 5\ V/\mu m$, now what is the new operating frequency and power consumption of the chip? (10 points)

Question 2

The minimum and maximum delays through the logic are annotated on the figure, and the flip-flops have the following properties: t_{clk-q} = 150 *ps*, t_{setup} = 50 *ps*, and t_{hold} = 100 *ps*. You can assume that the clock has no jitter, but t_{skew1} and t_{skew2} can be either positive or negative.

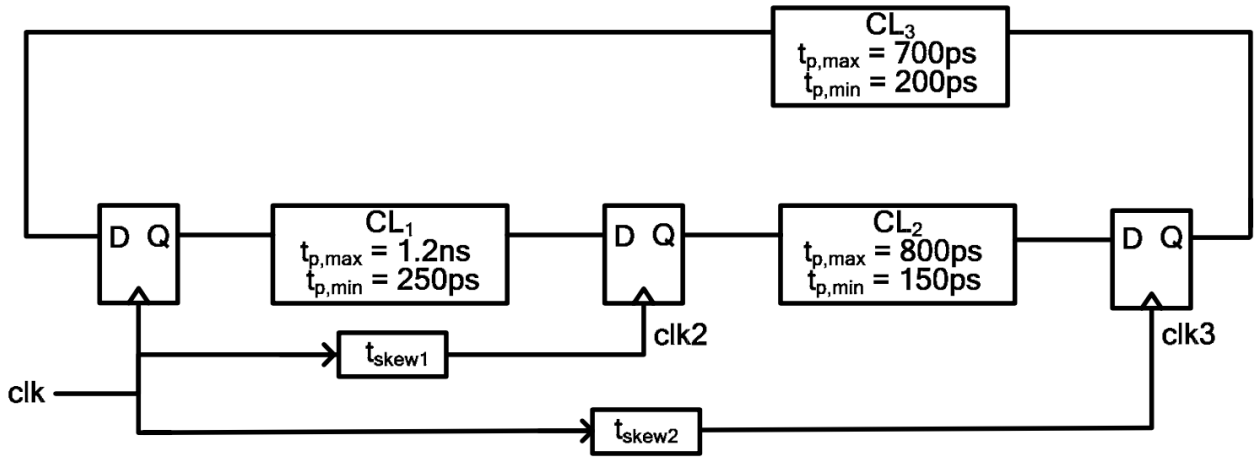


Figure 1

(A) What is the minimum clock cycle time if t_{skew1} = -100 *ps* and t_{skew2} = 50 *ps*? (7 points)

(B) Assuming t_{skew2} is fixed at 50 *ps*, how negative can t_{skew1} be before this pipeline fails a hold-time constraint? (7 points)

(C) If you could intentionally set the values of t_{skew1} and t_{skew2}, what values would you choose to minimize the cycle time of this pipeline? What would be the cycle time in this case? (6 points)

Question 3

(A) Shown below is a static CMOS implementation of a gate that computes the carry-out at a particular bit position. If the "P" signal that was fed into this gate was calculated using $A+B$ (instead of $A \oplus B$), would the output of this gate still be correct? Why or why not? (5 points)

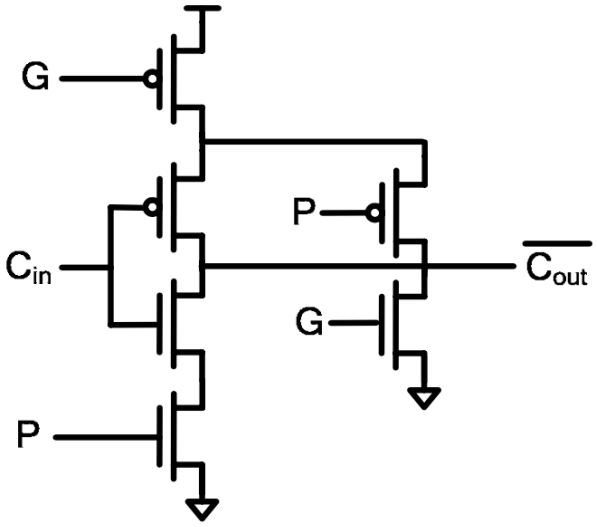


Figure 2

(B) Now let's look at the gate shown below that computes the carry-out for two-bit positions, but this time implemented with a Manchester carry chain. Is this gate guaranteed to function correctly if the P signals that are fed to this gate are calculated as $A + B$? Why or why not? (5 points)

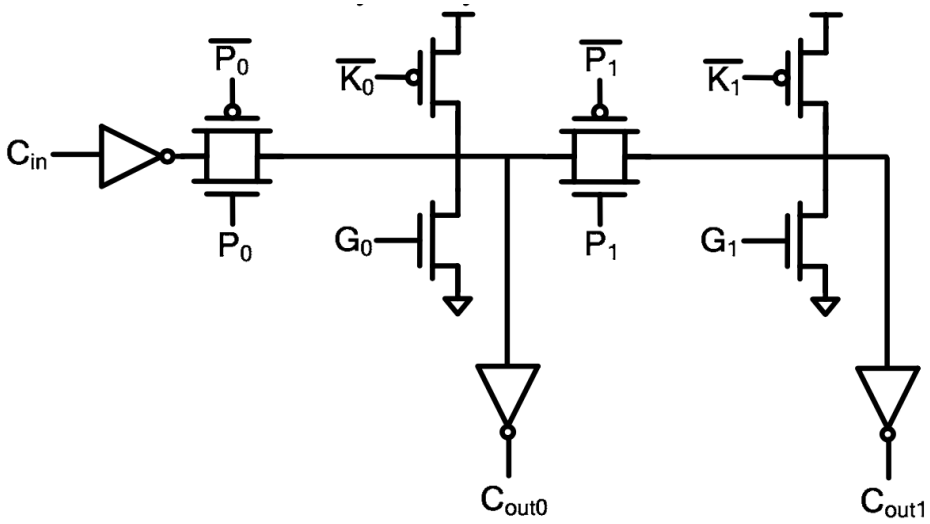


Figure 3

(C) Your colleague proposes to build a multiplier out of a ROM instead of out of full-add and half-add cells. How large of a ROM would you need to implement a 4-bit by 4-bit multiplier? Do you think this a good idea? Why or why not? Discuss your answer in terms of power, performance, and area. Be as detailed as possible. (10 points)

Question 4

(A) In your own words, describe the functionality of a carry-bypass adder, linear carry-select adder, and square-root carry-select adder. You can use diagrams to explain your answer. (10 points)

(B) In your own words describe the electro-migration problem in wires. (5 points)

(C) Compare and contrast a typical SRAM cell and a typical DRAM cell in terms of the number of transistors, volatility, and speed. (5 points)

Question 5

In this problem, we will be looking at a 256 x 256 SRAM, where each 6T cell is 2 μm wide and 1.5 μm tall. All of the devices in the SRAM cell are minimum length (L = 0.1 μm), and you can assume that both the NMOS access transistors and the NMOS pull-down transistors are 0.25 μm wide and that the PMOS transistors are 0.12 μm wide. The bitline wires are 0.2 μm wide. You can also assume that V_{DD} = 1.2 V, $C_G = C_D$ = 1 fF/μm, and that the wires have the following characteristics: C_{pp} = 100 aF/μm^2 , C_{fr} = 50 aF/μm/edge, and Rsq,w = 0.1 Ω/sq.

1. What is the total capacitance loading each bit line in this memory? (7 points)

2. If we were reading to or writing from the SRAM array, what is the worst-case RC delay due to the bit line wire? You can assume that the input is a ramp (i.e., $t_p = t_{Elmore}$). (7 points)

3. If the V_{DD} of the SRAM was raised to 1.3 V, would the RC delay of the bit line wire increase, decrease, or stay the same? You don't need to do any calculations, but you do need to explain your answer. (6 points)

Question 6

(A) Assume we have a 128 x 128 SRAM cell array (each wordline drives 128 cells and each bitline has 128 cells on it). In the SRAM cell, the width of both the access NMOS transistors is 120 *nm*, the width of the pullup PMOS transistors is 120 *nm*, and the width of the pulldown NMOS transistors is 240 *nm*. The SRAM cell's layout is 2 μm x 2.5 μm, and both wordline and bitline are 0.1 μm wide. You can assume $C_G = C_D = 2\ fF/\mu m$, $R_{sq,NMOS}$ = 10 $K\Omega$/sq, $R_{sq,PMOS}$ = 20 $K\Omega$/sq, and that for the wordline and bitline is R_w = 0.1 Ω/sq and $C_w = 0.2\ fF/\mu m$.

(a) Assuming this SRAM works at 1 GHz and V_{DD}=1.2 V, how much dynamic power is consumed due to the capacitance of the wordlines and bitlines during the read operation? Note that here you need to include the capacitance of the wire as well as the parasitic capacitance of the transistors connected to the wire. (10 points)

(b) Assuming that the final gate driving the wordline is always an inverter sized to have a fanout of four, how many SRAM cells can you place on each wordline before the delay caused by the resistance of the wordline wire is equal to the delay caused by the resistance of the inverter that drives the wordline? (10 points)

Bonus question
Prof Joshi is big into sports. Name five indoor games that he can play this winter. (5 points)